MIL-C-60983C (PA)
16 January 1975
SUPERSEDING
MIL-C-60983B (MU)
24 August 1970

MILITARY SPECIFICATION

CARTRIDGE, 40MM, HE, M406

LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of cartridge designated as Cartridge, 40MM, HE, M406.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	– Preservation, Methods of
MIL-P-20444	– Primer, M42
MIL-A-48078	– Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

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MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests For

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKING DRAWINGS

8833702	- Loading Assembly
8835947	- Projectile Assembly
	- Cartridge, 40MM, HE, M406
8844609	- Cartridge Case Loading Assembly M118
8835104	- Box, Wirebound, Packing, Ammunition for Cartridge, 40MM
8835105	- Box, Fiber, Packing, Ammunition for Cartridge, 40MM

INSPECTION EQUIPMENT DRAWINGS

8827889	- Chamber Gage	9202770 (Alternate)
8827890	- Alignment Check	9202783 (Alternate)
8827895	- Limit Checks	

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)").

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Cartridge. -The cartridge shall comply with all requirements specified on Drawing (Dwg.) 8835950, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Loading of high explosive.-

3.3.1 The ball and skirt assembly just before loading shall be inspected for presence of primer paint and any foreign matter. Ball and skirt assemblies with the interior wall in-completely coated or not dry shall not be loaded.

3.3.2 Solid explosive shall not be present before loading or added during pouring or cooling.

3.3.3 The fuze Seating surface shall be free of caked explosive. A chemical cleaner, if used, shall not contaminate the explosive.

3.3.4 Clean scrap produced in regular operations may be remelted and used provided it complies with the grade of explosive specified except for form and viscosity. Clean scrap is defined as scrap formed in kettles, pails, riser, and loading machines used in pouring, casting and drilling operations.

3.4 Air pressure. -The cartridge assembly shall withstand an air pressure of three (3) pounds-per-square-inch (psi) for five (5) seconds minimum, without a drop in pressure.

3.5 Primer Functioning. -The primer shall function without evidence of squibbing or hangfire. Para. 6.6 of MIL-P-20444B, Amendment 3, contains definitions of functioning failures.

3.6 X-ray examination of ballistic samples. -Prior to forwarding the cartridges to the proving ground for ballistic testing, as specified in 4.4.3.10 and 4.4.3.11, they shall be subjected to x-ray examination.

3.7 X-ray examination of fuze assemblies prior to assembling to ball and skirt assembly. -The fuze assembly shall be x-rayed for improperly assembled, missing parts, armed or partially armed (as evidenced by the detonator moving any degree from the safe position).

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3.8 Production control check on loaded M118 cartridge cases.- There shall be no evidence of a primer malfunction.

3.9 Functioning.-

3.9.1 Cartridge. -The cartridge shall function satisfactorily and the projectile shall have a mean velocity of 247 plus or minus (±) 7 feet-per-second and a standard deviation not exceeding 4.5 feet-per-second.

3.9.2 Safety. -There shall be no premature burst (See 6.8) or evidence thereof in any of the test firings.

3.10 Workmanship. -All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

3.11 Submission of results of contractor-conducted examinations and tests. -The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. -Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 (PA) shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. -The following types of inspection shall be conducted on this item:

a. Quality Conformance Inspection

4.3 First Article Inspection.-
Not applicable to this specification.

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation. -Inspection lots shall comply with the lot formation provisions of MIL-A-48078 (PA). In addition, inspection lots of cartridges shall contain:

a. Cartridge case metal parts from one interfix lot number from one manufacturer.
b. Ball and skirt assemblies from one interfix lot number from one manufacturer.
c. Composition B from one interfix lot number from one manufacturer.
d. Fuzes from no more than two (2) consecutively produced serial lots from one manufacturer.
e. Primers from one lot number from one manufacturer.
f. Propellant from one lot number from one manufacturer.

4.4.2 Examination. -See MIL-A-48078 (PA).

a Sampling Plans. -Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60983C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Cartridge Case Loading Assembly prior to assembling to projectile and prior to assembling base plug and primer		8844609
			NEXT HIGHER ASSEMBLY 8835950

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Powder charge cup missing		100%	3.2	Gage
2	Spilled powder from cartridge case		100%	3.2	Visual
3	Propellant charge weight obviously incorrect (See 6.11)		100%	3.2	Visual
Special	None defined				
Major A					
101	Propellant charge weight obviously incorrect (See 6.11)		100%	3.2	Visual
Major B	None defined				
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-60983C (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cartridge case loading assembly, prior to assembling to projectile	1 OF 1	8844609
			NEXT HIGHER ASSEMBLY 8835950

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Primer above flush with respect to base of case		100%	3.2	Gage
Special *a.	Vent hole blocked or missing		100%	3.2	Automatic Gage or Visual/Equipment
Major A 101	Base plug above flush		0.40%	3.2	Gage
102	Depth to base plug, max.		0.40%	3.2	Gage
103	Primer above flush or more than .005 below with respect to base plug		0.40%	3.2	Gage
104	Primer damaged		0.40%	3.2	Visual
Major B	None defined				
Minor 201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

*Prior to case loading

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60983C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Loading assembly, prior to loading				8833702
					NEXT HIGHER ASSEMBLY 8835947
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B 131	Specific gravity of charge	#	##	3.2	4.5.8
Minor 201	Foreign matter on interior surface		0.65%	3.3.1 3.10	Visual

Two (2) sample cup assemblies each four (4) hours production.
If any sample fails the four (4) hours production represented by the sample shall be rejected.

NOTES:

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8

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Loading Assembly		8833702
			NEXT HIGHER ASSEMBLY 8835947

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B 131	Excessive cavitation on bottom of fuze cavity		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Projectile, prior to assembling fuze and prior to assembling sealing compound on fuze threads				8835947
					NEXT HIGHER ASSEMBLY 8835950
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B					
131	Distance from seating surface on fuze to end of booster cup, max.		0.40%	3.2	Gage
132	Gap from fuze threads to booster cup missing		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-C-60983C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8835947
4.4.2.6	Projectile prior to assembling ogive				NEXT HIGHER ASSEMBLY 8835950
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A 101	Fuze torque failure		0.40%	3.2	Test
Major B 131	Length from curved surface to top of fuze		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60983C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Projectile		8835947
			NEXT HIGHER ASSEMBLY 8835950

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A 101	Weight incorrect		0.40%	3.2	Balance
Major B	None defined				
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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12

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60983C (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Cartridge	1 OF 1	8835105
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD	
						Alternate
Critical 1	Chamber gage failure (after complete cure)		100%	3.2	8827889 8827890 8827895	9202770 9202783
Special	None defined					
Major A 101 102	Security of projectile Air pressure	80 200	1.50%* 1.50%	3.2 3.2	4.5.3 4.5.6	
Major B	None defined					
Minor 201 202 203	Total length Marking missing Evidence of poor workmanship		0.65% 0.65% 0.65%	3.2 3.2 3.10	Gage Visual Visual	

* In addition, if any cartridge tested has a value of less than fifty (50) pounds or any cartridge case or projectile shows any evidence of uncured sealant after completion of this test, the lot shall be rejected.

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Unsealed fiber box				8835105
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B					
131	Improper number of cartridges		0.40%	3.2	Visual
132	Separator missing		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60983C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Sealed fiber box, prior to sealing bag		8835105
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B					
131	Heat seal test of packing seals	(a)	(a)	3.2	4.5.7
132	Container damaged		0.40%	3.2	Visual
133	Corners not blunted		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual

(a) Sampling and rejection shall be in accordance with MIL-P-116.

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-60983C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Unsealed wooden packing box				8835104
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B					
131	Marking of bag misleading or unidentifiable		0.40%	3.2	Visual
132	Bag torn or perforated		0.40%	3.2	Visual
133	Bag improperly sealed		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60983C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Sealed wooden packing box				8835104
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special	None defined				
Major A	None defined				
Major B					
131	Box damaged		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Car seal missing or improperly positioned		0.65%	3.2	Visual
203	Marking misleading or unidentifiable		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing.-

4.4.3.1 Check test for deterioration of primers (See MIL-P 20444). -If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two (2) years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-P-20444. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into cartridge (See 6.6).

4.4.3.2 Holding fixture vent check. -The cartridge case holding fixture used in the base plug insertion operation shall have its air vents check per shift for any obstruction. If a vent is found to be closed or clogged, the vent, shall be cleared and the previous hour's production shall be x-rayed 100 percent to ensure that the propellant charge cups have not been damaged. Cartridge cases found to have damaged charge cups shall be classed defective and removed from the lot.

4.4.3.3 Propellant weight of cartridge case. -The propellant weight shall be determined and then check weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing or determination, using a different balance from that used to make the original weighing and if performed manually, shall be performed by another operator. Any charge which falls to comply with the requirement specified on Dwg. 8844609 shall be classified defective and removed from the lot.

4.4.3.4 Propellant weight of cartridge case, Alternate method.- At the start of production, three hundred eighty (380) consecutive samples per station, per machine shall be selected and weighed 100 percent. All samples shall comply with the applicable drawing requirement. If all samples comply with the drawing requirement, five (5) samples, per station, per every two (2) hours production shall be selected and the average weight of the five (5) samples must meet the assessed mean propellant weight within ± 4 milligrams. If the average weight fails the requirement, correction will be made and three hundred eighty (380) consecutive samples weighed from that station. If all meet the requirement the sampling every two (2) hours will be resumed.

4.4.3.5 X-ray examination of fuze assemblies prior to assembling to ball and skirt assembly(See Para. 3.7). -Each fuze assembly, prior to assembling to ball and skirt assembly, shall be subjected to this test. If any fuze assembly shows evidence of being improperly assembled, having missing parts, being armed or partially armed, it shall be classed defective and removed from the lot.

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4.4.3.6 Production control check on loaded M118 Cartridge Case (See Para. 3.8). -If (5) loaded M118 cases shall be randomly selected from every two (2) hours production from each primer assembly machine. The loaded M118 cases produced by each primer assembly machine shall be kept segregated and identified. They shall not be used until this test has been performed. If any sample fails to comply with the applicable requirement, the two hours production represented by the sample shall be rejected.

4.4.3.7 Security of crimp of base plug on cartridge case assembly (See Dwg. 8844609). -Twenty (20) each of cartridge case, base plug, and powder charge cup less propellant shall be selected from each lot and subjected to this test. If the crimped area of one or more assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected.

4.4.3.8 Primer functioning (See Para. 3.5). -Sampling shall be in accordance with Paragraph 4.2.3.2.2 of MIL-P-2044B, except that the sample size shall be eight hundred (800) primers from each lot. If one or more primers fall to comply with the applicable requirement, the lot shall be rejected. Primers not originally purchased to this requirement shall be tested for the requirement of Paragraph 3.5.

4.4.3.9 Security of crimp of ogive of projectile (See Dwg. 8835947 (when applicable). -Twenty (20) each of ogive and empty ball and skirt assemblies shall be selected from each lot. The ogive shall be cemented and crimped in place and subjected to this test. If the ogive of two (2) or more assemblies fail to comply with the minimum requirement specified on the applicable drawing, the lot shall be rejected.

4.4.3.10 Functioning (See para. 3.9). -Beginning with the first lot produced and continuing until 3 acceptable lots have been accepted under the rejection criteria specified in 4.4.3.10.1, the Government inspector shall select 288 cartridges from each lot for this test. Ninety six (96) cartridges shall be fired from each of three (3) randomly selected launchers.

4.4.3.10.1 Rejection. -The lot shall be rejected if:

a. 12 or more fail high order detonation at initial impact (see 6.9). (major defect)
b. Any primer blow backs (as evidenced by perforation of the cup and extrusion of the primer around firing pin causing difficulty in opening weapon). (Major defect)
c. Any premature burst occurs. (Critical defect)
d. Mean velocity and/or standard deviation fail to meet the applicable requirement. (Major defect)
e. Any primer misfires. (Major defect)

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f. (1) 30 or more cartridge cases cannot be ejected from the launcher by spring operation of the extractor. (Major defect)
(2) 1 or more cartridge cases cannot be ejected from the launcher by hand operation of the extractor. (Major defect)
g. Any projectile sticks in gun bore or falls to attain 60 percent of the mean velocity of the lot sample. (A projectile on which a velocity reading was not obtained due to low velocity shall be considered a failure). (Critical defect)
h. 4 or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug or cartridge case base). (Major defect)
i. Any metal part separation occurs prior to impact. (Critical defect)

4.4.3.10.2 Initial production test plan, (informational test only). -Beginning with the first lot produced by each producer and continuing until that producer has submitted three (3) acceptable lots that are accepted in accordance with Paragraph 4.4.3.10 an extra sample of ninety (90) rounds shall be selected from each lot and preconditioned and tested as follows:

a. The ninety (90) rounds shall be packaged in their regular shipping containers and subjected to transporation vibration environment in accordance with MIL-STD-331, Test No. 104.
b. After transporation vibration, the rounds shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.
c. After temperature and humidity cycle, the rounds shall be temperature conditioned and fired as follows:

30 rounds + 165 degrees F ± 5 degrees F.
30 rounds ambient (between + 40 and + 110 degrees F).
30 rounds -65 degrees F ± 5 degrees F.

d. Fire the rounds in accordance with Paragraph 4.5.10 and with TECOM Material Test procedure, TECP 700-700 Volume 4. Test results, failure analysis reports, and other information as requested shall be forwarded to Picatinny Arsenal, Attn: SARPA-QA-A-S for evaluation.

4.4.3.11 Functioning (See para. 3.9) - Regular Production.- After 3 acceptable lots have met the criteria of 4.4.10.1 the Government inspector shall select 144 cartridges from each lot for this test. Forty eight (48) cartridges shall be fired from each of three (3) randomly selected launchers. If 2 or more consecutive lots fail the criteria of 4.4.3.11.1, the lots thereafter shall revert to the criteria of 4.4.3.10.

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4.4.3.11.1 Rejection. -The lot shall be rejected if:

a. 7 or more fall high order detonation of initial impact (See 6.9) (Major defect)
b. Any primer blow backs (as evidenced by perforation of the cup and extrusion of the primer cup around firing pin causing difficulty in opening weapon) (Major defect)
C. Any premature burst occurs (Critical defect)
d. Mean velocity and/or standard deviation fall to meet the applicable requirement (Major defect)
e. Any primer misfires (Major defect)
f. (1) 10 or more cartridge cases cannot be ejected from the launcher by spring operation of the extractor. (Major defect)
(2) 1 or more cartridge cases cannot be ejected from the launcher by hand operation of the extractor (Major defect)
g. Any projectile sticks in gun bore or falls to attain 60 percent of the mean uncorrected velocity of the lot sample. (A projectile on which a velocity reading was not obtained due to low velocity shall be considered a failure) (Critical defect)
h. 3 or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug or cartridge case base) (Major defect)
i. Any metal part separation occurs prior to impact. (Critical defect)

4.4.3.12 X-ray examination of ballistic samples (see 3.6).- Prior to forwarding the cartridges to the proving ground for ballistic testing, as specified in 4.4.3.10 and 4.4.3.11, they shall be subjected to x-ray examination. The cartridges shall be x-rayed for critical and major defects. If any critical defect is found, the lot shall be rejected including the ballistic samples. If any major defect is found it should be noted and replaced with a good sample and the samples forwarded to the proving ground.

4.4.4 Inspection equipment. -The inspection equipment required to Perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables. starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-4876 (PA) and 6.3 herein.

4.5 Test methods and procedures.-

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4.5.1 X-ray examination of fuze assembly prior to assembling the ball and skirt assembly. -Improperly assembled, missing parts, armed or partially armed fuze assemblies shall be determined by x-ray equipment or any other method satisfactory to the Contracting Officer. This test is a non-destructive test. Parts so tested may be returned to the lot.

NOTE: This test shall be conducted using the side view. The fuze shall be orientated so that the indent on the fuze cover is closest to the source of the x-ray beam. The following minimum visual verifications are required to assure a safe fuze:

a. Setback pin visible through the rotor gear.
b. Detonator is located over setback pin.
c. Centrifugal lock present and engaged in starwheel
d. Presence of verge.

4.5.2 Production control check on loaded M118 cartridge cases. -The loaded M118 cartridge case shall be fired from an approved launcher placed on an approved mount. Observation shall be made for primer functioning. This test is a destructive tests. Parts so tested shall not be returned to the lot. NOTE: No projectile is required for this test.

4.5.3 Security of projectile. -The cartridge shall be placed in an approved fixture and the axial force specified on Dwg. 8835950 shall be applied. Projectile must separate from the cartridge case in the range specified on Dwg. 8835950. This test is a non-destructive test. Parts so tested may be returned to the lot. Stake shall be removed from the loaded case prior to returning to the lot.

4.5.4 Security of crimp of base plug on cartridge case assembly.- The assembly shall be placed in an approved fixture and the minimum unseating force specified on Dwg. 8844609 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.5 Security of crimp of ogive of projectile (When applicable).- The assembly shall be placed in an approved fixture and the minimum unseating force specified on Dwg. 8835947 shall be applied until failure of the crimp area of the ogive occurs. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.6 Air Pressure. -The cartridge shall be placed in an approved fixture and a measured quantity of air applied to produce the required air pressure. This test is a non-destructive test. Parts so tested may be returned to the lot.

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4.5.7 Heat seal test of packing seals. -This test shall be performed in accordance with MIL-P-116. Seals so tested shall not be returned to the lot.

4.5.8 Specific gravity of the charge of ball and skirt loading assembly. -The specific gravity of the composition B shall be determined as follows:

Weigh the empty ball and skirt assembly in grams and record as We. Fill the empty assembly with water at 75 degrees ± 2 degrees F., weigh in grams and record the weight as Ww. Determine the volume by subtracting We from Ww.

Load the assembly with Composition B, drill the fuze well cavity, weigh in grams and record the weight as W1. Determine the charge weight by subtracting We from W1. Calculate the specific gravity as follows:

$$\text{Specific gravity in grams/CM.} = \frac{W1 - We}{Ww - We} - 3.30$$

4.5.9 X-ray examination of ballistic samples. -The cartridge, prior to forwarding to the proving ground, shall be subjected to x-ray examination using approved x-ray equipment or any other method satisfactory to the contracting Officer. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.5.10 Functioning. -This test shall be conducted at a Government owned proving ground. The cartridge shall be fired from an approved launcher-placed on an approved mount to impact at 200 meters minimum from the launcher on a six inch thick loose gravel pad. The gravel used shall have a 1/2 inch maximum size. The velocity shall be measured at 20 feet from the muzzle of the launcher using two sets of screens, each set of screens having an electronic counter. One set of screen shall be placed 14 and 26 feet from the launcher and the other set 15 and 25 feet from the launcher. NOTE: To insure that the chronographs display accurate time intervals the ratio of the two times shall not differ from 1.200 by more than plus or minus 0.005. If the ratio differs from 1.200 by more than plus or minus 0.005, the velocity results for that round shall be discarded and excluded from the calculation of mean velocity and velocity standard deviation. (No more than eight (8) rounds in a sample size of 144, or twelve (12) rounds in a sample size of 288 may be disregarded. If the applicable number of improper ratios is exceeded, additional round(s) must be fired) Observation shall be made and results recorded for functioning, mean velocity and standard deviation (See 6.7).

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4.5.10.2 Test Validity. -If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.5.11 Primer functioning. -Each primer of the 800 samples shall be tested as specified in MIL-P-20444B, paragraph 4.3.2.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A. -The cartridge shall be preserved and packaged in accordance with Dwg. 8835105.

5.2 Packing.-

5.2.1 Level A. -The cartridge, preserved and packaged as specified in shall be packed in accordance with Dwg. 8835104.

5.3 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

5.4 Marking.-Marking shall be in accordance with Dwg. 8835104, and 8835105, and Code of Federal Regulations 49 Title, Parts 0-190.

6. NOTES

6.1 Intended Use. -This specification covers the loading, assembling and packing for Cartridge, 40MM, HE, M406.

6.2 Ordering Data. -See MIL-A-48078 (PA).

6.3 Submission of Inspection Equipment Designs for Approval.- See MIL-A-18078 (PA). Submit equipment designs as required to Commander, Picatinny Arsenal, Attn: SARPA-QA-T, Dover, New Jersey 07801.

6.4 Submission of results of contractor-conducted examinations and tests. -Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, Picatinny Arsenal, Attn: SARPA-QA-A-S.

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6.5 Submission of test data. -In addition to the normal distribution of records, when the cartridges are procured by the U.S. Army Munitions Command one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, Picatinny Arsenal, Attn: SARPA-QA-A-S.

6.6 Cost of check test. -The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the primer assemblies or to the contractor assembling the primer assemblies into the cartridge and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness In handling, storage, etc., permitted while the primer assembly lot was under the jurisdiction of either contractor.

6.7 Standard deviation shall be calculated from the following formula or other approved formula:

$$S = \sqrt{\frac{\Sigma (X - \bar{X})^2}{n - 1}}$$

where:

X = each individual value

$\bar{X}$ = Sample arithmetic means $= \frac{\Sigma X}{n}$

n = sample size

$\Sigma (X - \bar{X})^2$ = the sum of the squares of the differences between the sample mean and each individual value

n - 1 = one less than the sample

6.7.1 For a faster and easier method of arriving at the standard deviation, the following alternate method may be used.

$$S = \sqrt{\frac{n \Sigma x \cdot - (\Sigma X)^2}{n (n - 1)}}$$

6.8 Premature burst. -A premature burst is considered to have occurred if the round functions in gun bore or in flight.

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6.9 High order functioning. -Evidence of high-order functioning shall be proper disintegration of the ball as determined by the proof director.

6.10 Combining of proving ground tests. -When the contractor for the cartridge is also the contractor for one or more of the components thereof the proving ground tests of the contractor may be combined with the proving ground tests of the cartridge, to save expense, upon agreement between the procuring activity and the contractor. In cases where the cartridge specification does not cover all of the proving ground tests specified for the component, the addition tests specified in the component specification shall be conducted.

6.11 If any cartridge case, prior to insertion of the base plug and primer, is suspected of containing an underweight or over-weight propellant charge, based on visual inspection, It will be removed from the lot and weighed on suitable scales. Any cartridge case found with less than 75% or more than 125% of the assessed propellant load will be classified as a Critical defect and removed from the lot.

Any cartridge case found to have a propellant weight out of drawing tolerance but within ± 25 of the assessed propellant load will be classified as a Major defect and removed from the lot.

6.12 Table shall be used at the loading plants for powder assessment.

6.13 Proving ground test summary:

Test	Sample Size		Requirements
Functioning	First 3 lots	- 288	See 3.9 and 4.4.3.10.1
	REG PROD	144	See 3.9 and 4.4.3.11.1

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1310-A071

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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